

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460395.851 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.052
Transit depth (Rp/Rs)^2	3.39 +/- 1.93 [%]
Orbital Inclination (inc)	84.54 +/- 1.96
Airmass coefficient 1 (a1)	0.945 +/- 0.035
Airmass coefficient 2 (a2)	0.046 +/- 0.055
Scatter in the residuals of the lightcurve fit is	3.68 %
Best Comparison Star	#2 - [186, 324]
Optimal Aperture	3.36
Optimal Annulus	5.51
Transit Duration (day)	0.05 +/- 0.045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.052 vs 0.1457 (deviation 0.74σ)
Duration fit vs published	0.05 d vs 0.12 d (deviation 1.56σ)
Mid-time O–C	3.2 min
Depth SNR	3.5
Residual scatter	3.68 %

Light curve

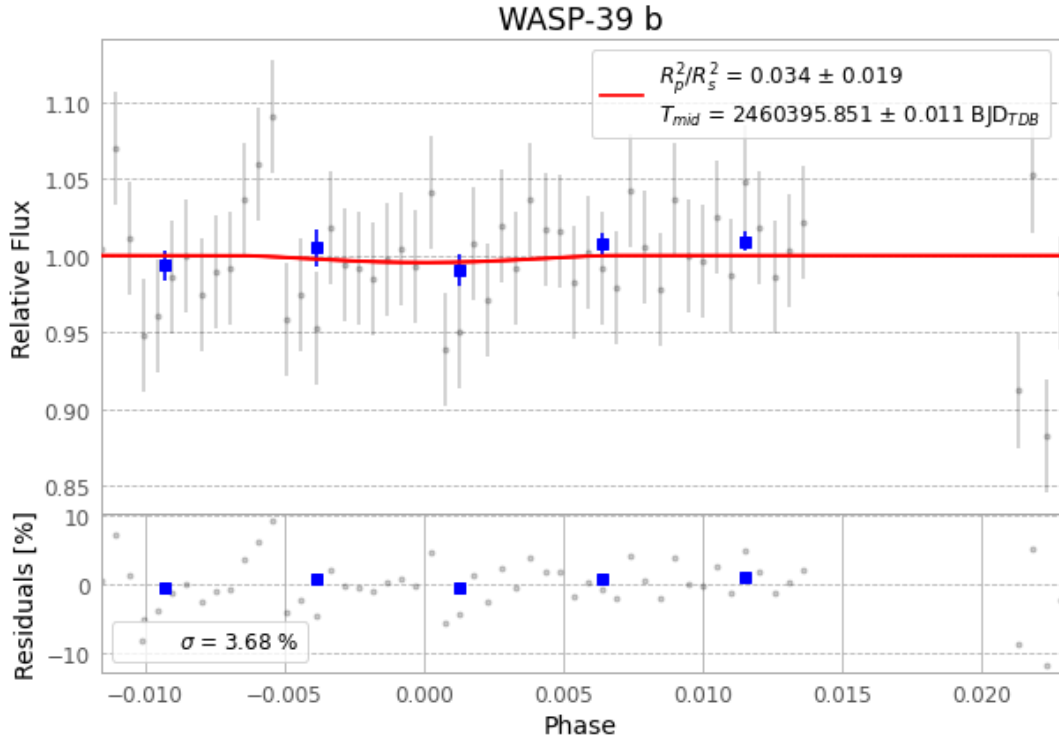


Figure 1 — FinalLightCurve_WASP-39 b_2024-03-26.png (SHA-256 12db4ac52de11712...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [413, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d2113a9789237562...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), WASP-39240326070916.FITS → WASP-39240326103615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-240326.log (11a270206abb...), FinalLightCurve_WASP-39 b_2024-03-26.png (12db4ac52de1...), FinalLightCurve_WASP-39 b_2024-03-26.csv (46b79ac0a86f...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 22:55Z.